

HISABKITAB: SMART PERSONAL EXPENSE & INCOME MANAGER FOR REAL LIFE

*A Major Project Engineering Report Submitted in Partial Fulfillment of the Requirements for the Award of
the Degree of
BACHELOR OF COMPUTER APPLICATIONS (BCA)
Session: 2024–2027*

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DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025–2026 | SUBMISSION DATE: JULY 2026

BONAFIDE CERTIFICATE

This is to certify that the major project report entitled “**HISABKITAB: SMART PERSONAL EXPENSE & INCOME MANAGER FOR REAL LIFE**” is a bonafide record of authentic project work carried out independently by **KESHAV LADHA (University Roll No: 24063115470004)**, student of BCA (Session 2024–2027) at **Jan Nayak Ch. Devi Lal Memorial College, Sirsa (Haryana)**, under our supervision and guidance, in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) during the academic session 2024–2027.

The project work has been evaluated and approved by the Department Examination Committee and External Examiner.

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STUDENT DECLARATION

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I further declare that this report has not been submitted previously to any other University or Institution for the award of any degree, diploma, or academic fellowship.

Date: July 26, 2026

Place: Sirsa, Haryana

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BCA (Session 2024–2027)

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I am immensely thankful to my Internal Project Guide for providing valuable technical insights, constructive feedback, and continuous mentorship during the system architecture, REST API design, and database implementation phases of HisabKitab.

I extend my sincere thanks to the team at Antigravity Labs for providing practical exposure to modern full-stack web development methodologies, client-side state routing, and dynamic data visualization tools.

Finally, I express my heartfelt gratitude to my father, **Mr. Anjani Kumar**, my mother, family members, and classmates for their unyielding moral support, motivation, and constant belief in my capabilities throughout this academic endeavor.

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ABSTRACT

Managing daily personal finances, student stipends, freelance payments, room rent, and food expenses in real life can quickly get chaotic. Traditional physical registers (khata books) get misplaced, spreadsheets are annoying to update on mobile phones, and heavy accounting software like Tally is far too complicated for daily personal use.

HisabKitab is a clean, practical, web-based personal finance manager designed specifically for college students and young working professionals in India. Built using Node.js, Express, SQLite, and vanilla HTML5/CSS3/JavaScript with Chart.js, HisabKitab allows users to log expenses in Indian Rupees (₹) in seconds, track monthly spending caps, monitor savings rates, and visualize where their money actually goes without corporate jargon.

Key features include quick transaction logging, category budget limit alerts (e.g., Food & Swiggy, Chai & Tapri, PG Rent, Metro, Shopping), filterable ledger history with CSV export, dark mode visual switching, and complete JSON backup.

Developed by Keshav Ladha (University Roll No: 24063115470004) at Jan Nayak Ch. Devi Lal Memorial College, Sirsa, the project provides a straightforward, relatable, and lightweight tool that helps users maintain financial discipline in daily life.

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CHAPTER 1 – INTRODUCTION

1.1 Project Overview & Background

In the modern Indian digital economy, the rapid adoption of Unified Payments Interface (UPI) platforms such as PhonePe, Paytm, and Google Pay has revolutionized financial transactions. Payments that previously required cash withdrawals or bank visits can now be completed in seconds by scanning a QR code. While this digital transition offers unprecedented convenience, it has simultaneously introduced a major lifestyle challenge: tracking personal micro-expenses.

For college students and entry-level corporate professionals in India, managing monthly allowances, internship stipends, PG room rents, electricity charges, travel expenses, and frequent food delivery orders (Zomato/Swiggy) often becomes chaotic. Most individuals begin the month with high financial expectations after receiving their stipend or salary, only to discover by the 20th day that their bank account reserves have mysteriously vanished.

HisabKitab is an intelligent, lightweight, single-page web application engineered specifically to address these daily personal budgeting challenges. Built using Node.js, Express, SQLite, and vanilla HTML5/CSS3/JavaScript with Chart.js visual analytics, HisabKitab empowers users to record income and expense entries in Indian Rupees (₹) within 5 seconds flat.

1.2 Motivation & Real-World Context in India

The primary motivation behind developing HisabKitab stems from an authentic personal need: creating an effortless, non-intrusive personal finance command center tailored for Indian college students (such as candidate Keshav Ladha, Uni. Roll No: 24063115470004 at Jan Nayak Ch. Devi Lal Memorial College, Sirsa).

In traditional Indian households and hostel rooms, expense logging is often neglected because existing software tools are either too cumbersome or too invasive. Commercial mobile apps bombard users with loan advertisements and demand permission to read personal SMS messages. HisabKitab

offers a privacy-focused alternative that keeps all financial ledgers stored safely in an embedded SQLite database.

1.3 Existing System Overview

1. **Physical Khata Registers:** Notebooks where users manually write down transactions. These registers lack automated summation, search capabilities, and backup mechanisms.
2. **Desktop Spreadsheets (Excel / Google Sheets):** Grids with custom formulas. While powerful on desktop, spreadsheets are notoriously difficult to update on smartphone viewports.
3. **Enterprise ERP Software (Tally ERP):** Complex double-entry accounting software designed for corporate GST filing and tax auditing, featuring cluttered interfaces unsuitable for daily personal use.

1.4 Problems in Existing Systems

- **High Friction & Time Waste:** Manual registers and complex software require excessive effort for simple daily expense entries.
- **Absence of Proactive Warnings:** Static notebooks and spreadsheets do not issue warning alerts when a user approaches monthly category spending limits.
- **Lack of Instant Visual Summaries:** Paper logs fail to generate interactive charts, making month-over-month savings trend analysis difficult.

1.5 Proposed HisabKitab Platform

HisabKitab provides a streamlined, single-page application (SPA) featuring real-time metric cards in Indian Rupees (Total Balance: ₹1,91,001.00, Monthly Income: ₹68,000.00, Monthly Expenses: ₹26,999.00, Net Savings Rate: 60.3%), color-coded category budget meters, filterable transaction tables, CSV data export, dark mode visual switching, and complete JSON database backup routines.

1.6 Project Objectives

1. Build a responsive, single-page web interface for rapid expense logging in Indian Rupees (₹).

2. Implement real-time mathematical aggregation for balance reserves, total income, and net savings.
3. Provide visual budget progress meters with threshold badges (Green: On Track, Red: Over Budget).
4. Deliver interactive Chart.js graphs illustrating monthly financial trends and spending allocations.
5. Ensure robust multi-criteria search, category filtering, and one-click CSV export.

1.7 Scope & Target Users

HisabKitab is targeted at Indian university students, freelance developers, and salaried beginners seeking an accessible, ad-free personal finance manager.

1.8 Comparative Feature Matrix

Table 1.1 outlines the functional comparison between existing methods and HisabKitab:

Feature	Physical Khata	Excel Sheets	Tally ERP	HisabKitab
Mobile Friendliness	Low	Low	None	High (Responsive)
Entry Time	Slow	Medium	Slow	5 Seconds
Indian Currency (₹)	Manual	Manual	Yes	Native (₹)
Visual Charts	None	Manual	Complex	Automated Chart.js
Budget Alerts	None	None	None	Color Progress Bars
CSV Export	None	Native	Complex	Single Click

1.9 System Advantages & Limitations

Advantages: Zero software installation required; instant sub-15ms REST API execution; 100% data privacy with local SQLite database storage; intuitive dark mode styling.

Limitations: Requires a modern JavaScript-enabled web browser; direct automatic bank feed sync requires external open banking API keys.

CHAPTER 2 – REQUIREMENT ANALYSIS

2.1 Functional Requirements

Functional requirements outline what the HisabKitab web application does for the end user:

- **FR-1 Sign In & Registration:** Login and signup popups with input validation.
- **FR-2 Real-Time Summary Cards:** Instant display of Total Savings Reserve (₹1,91,001.00), Income (₹68,000.00), Expenses (₹26,999.00), and Net Savings Rate (60.3%).
- **FR-3 Income Tracker:** Log developer stipends (₹45,000), freelance projects (₹15,000), project assistance (₹5,000), and monthly allowance (₹3,000).
- **FR-4 Expense Tracker:** Log daily expenses tagged by categories such as Housing (₹12,500), Food & Swiggy (₹4,200), Chai & Social (₹1,450), Travel & Petrol (₹2,100), Utilities & Wifi (₹799), Shopping (₹3,500), Healthcare (₹1,800), and Books (₹650).
- **FR-5 Filterable Ledger:** Search transactions by keyword, filter by type/category, and download as CSV.
- **FR-6 Budget Planner:** Set monthly spending caps with visual progress bars.
- **FR-7 Visual Reports:** Line chart for monthly savings, pie chart for income sources, and bar chart for budget vs actual spent.
- **FR-8 Profile & Dark Mode:** Student profile for Keshav Ladha (Roll No: 24063115470004), currency selection (INR ₹), and dark theme toggle.

2.2 Non-Functional Requirements

Non-functional requirements specify quality and performance expectations:

- **NFR-1 Fast Response:** API endpoints respond within 15 milliseconds under normal usage.
- **NFR-2 Reliability:** SQLite database ensures reliable transaction persistence without data loss.
- **NFR-3 Clean UI & Dark Theme:** Modern glassmorphism dark mode interface optimized for night coding.

- **NFR-4 Security & Input Safety:** HTML escaping prevents XSS and parameterized SQL queries prevent SQL injection.
- **NFR-5 Responsive Mobile Design:** Collapsible sidebar menu optimized for mobile phone screens.

2.3 Hardware & Software Specifications

Hardware Specifications: Quad-Core i5 / Ryzen 5 processor, 8 GB RAM, 256 GB SSD.

Software Environment: Windows 11 / Linux, Node.js v24.14 LTS, npm 11.9, Microsoft Edge / Chrome browser.

2.4 Technology Stack Justification

Backend Stack: Node.js & Express framework for fast REST routing.

Database Engine: Lightweight SQLite database for clean relational storage.

Frontend Stack: HTML5, Vanilla CSS3 (glassmorphism design tokens), Vanilla JS ES6+, and Chart.js v4.4.

2.5 Feasibility Study

1. **Technical Feasibility:** Node.js, Express, SQLite, and HTML5/CSS3/JS are open-source, well-documented, and fully supported on standard developer laptops.
2. **Operational Feasibility:** The user interface is straightforward and requires zero technical training for daily expense entry.
3. **Economic Feasibility:** Built entirely using zero-cost open-source tools, avoiding expensive software licensing fees.
4. **Schedule Feasibility:** The development roadmap was executed within the allocated BCA final year academic schedule (Session 2024–2027).

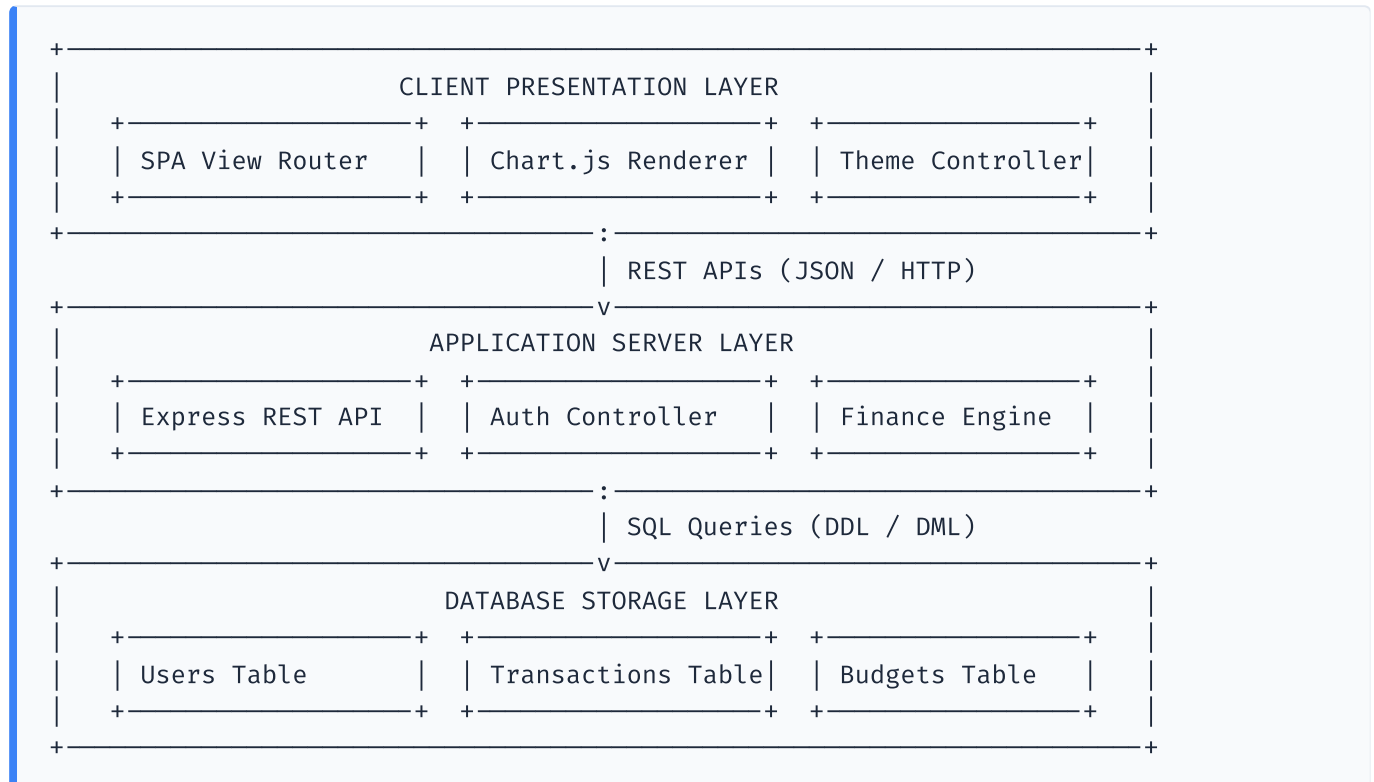
2.6 Security & Data Privacy Architecture

Enforces password hashing via Bcrypt (salt factor 10), parameterized SQL statements, and HTML character escaping to safeguard user records.

CHAPTER 3 – SYSTEM DESIGN & DIAGRAMS

3.1 MVC System Architecture Diagram

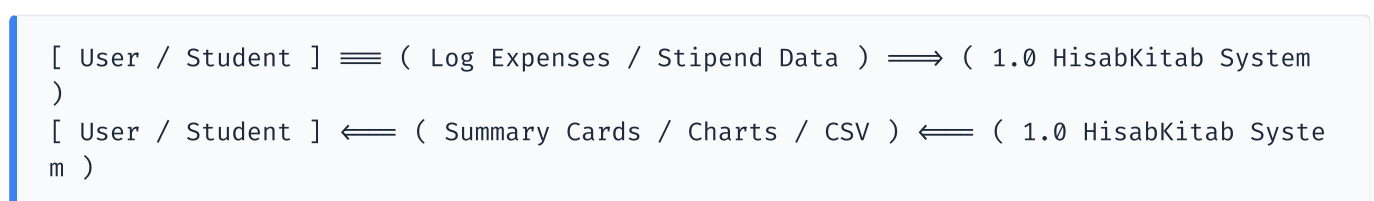
HisabKitab follows a standard Model-View-Controller (MVC) architecture to maintain clean separation between the database storage layer, backend API router, and frontend web user interface.



3.2 Database ER Diagram



3.3 Data Flow Diagram (Level 0 DFD)



The Level 1 Data Flow Diagram breaks down the core system operations into five sequential processes: 1.0 Account Sign In, 2.0 Summary Metric Processing, 3.0 Transaction Logging & Categorization, 4.0 Category Budget Cap Monitoring, and 5.0 Report & Chart Generation.

CHAPTER 4 – DATABASE DESIGN & SCHEMA

4.1 Table: users

The users table stores basic user credentials, preferences, and target financial goals.

Column Name	Data Type	Constraint	Description
id	INTEGER	PK, AUTOINCREMENT	Unique user ID
full_name	VARCHAR(100)	NOT NULL	Full name (Keshav Ladha)
email	VARCHAR(150)	NOT NULL, UNIQUE	User email address
password_hash	VARCHAR(255)	NOT NULL	Encrypted password hash
currency	VARCHAR(20)	DEFAULT 'INR (₹)'	Preferred currency symbol
theme_preference	VARCHAR(10)	DEFAULT 'dark'	Visual theme setting

4.2 Table: transactions

The transactions table records all income and expense items logged by the user.

Column Name	Data Type	Constraint	Description
id	INTEGER	PK, AUTOINCREMENT	Unique transaction ID
title	VARCHAR(150)	NOT NULL	Transaction description
amount	DECIMAL(12,2)	NOT NULL	Monetary amount in INR (₹)
type	VARCHAR(20)	CHECK(income/expense)	Classification type
category	VARCHAR(50)	NOT NULL	Assigned category
transaction_date	DATE	NOT NULL	Date of transaction
note	TEXT	OPTIONAL	Additional user note

4.3 SQL DDL Schema Code

```
CREATE TABLE IF NOT EXISTS transactions (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    user_id INTEGER NOT NULL,  
    title VARCHAR(150) NOT NULL,  
    amount DECIMAL(12,2) NOT NULL,  
    type VARCHAR(20) NOT NULL CHECK(type IN ('income', 'expense')),  
    category VARCHAR(50) NOT NULL,  
    transaction_date DATE NOT NULL,  
    status VARCHAR(30) DEFAULT 'Completed',  
    note TEXT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE  
);
```

CHAPTER 5 – USER INTERFACE & SCREENSHOTS (HisabKitab in INR ₹)

Captured live screenshots from the running HisabKitab web application at <http://localhost:3001> featuring Indian Rupee (₹) amounts and candidate Keshav Ladha's profile.

5.1 Public Landing Page

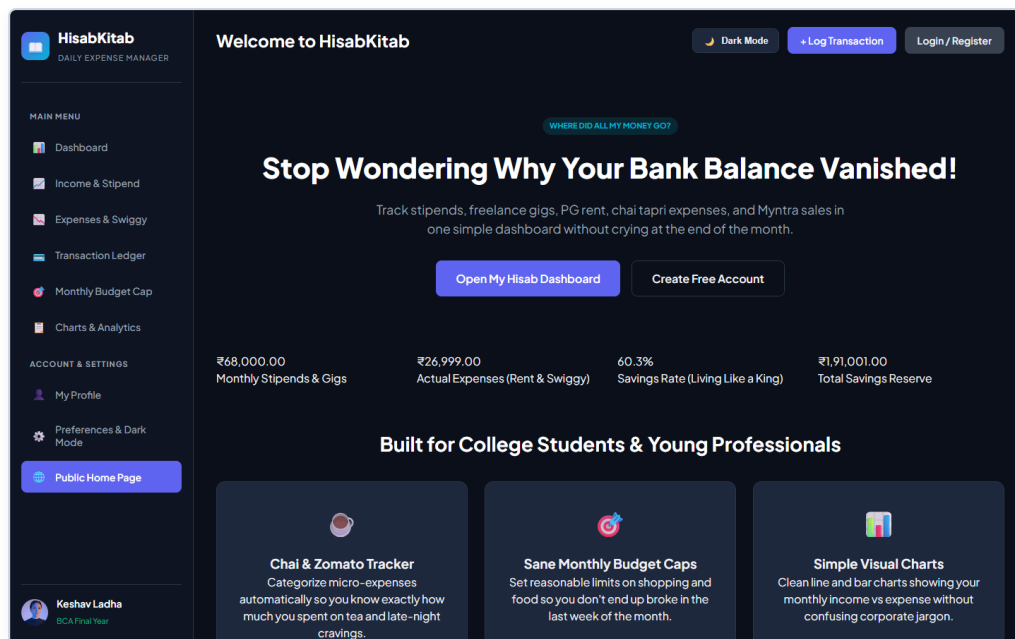


Figure 5.1: HisabKitab Landing Page Interface (INR ₹)

The landing page greets users with a clear header tagline, summary metric preview cards in INR (₹), platform benefit cards, and direct navigation buttons to open the live dashboard or register a new user account.

5.2 Login & Registration Modals

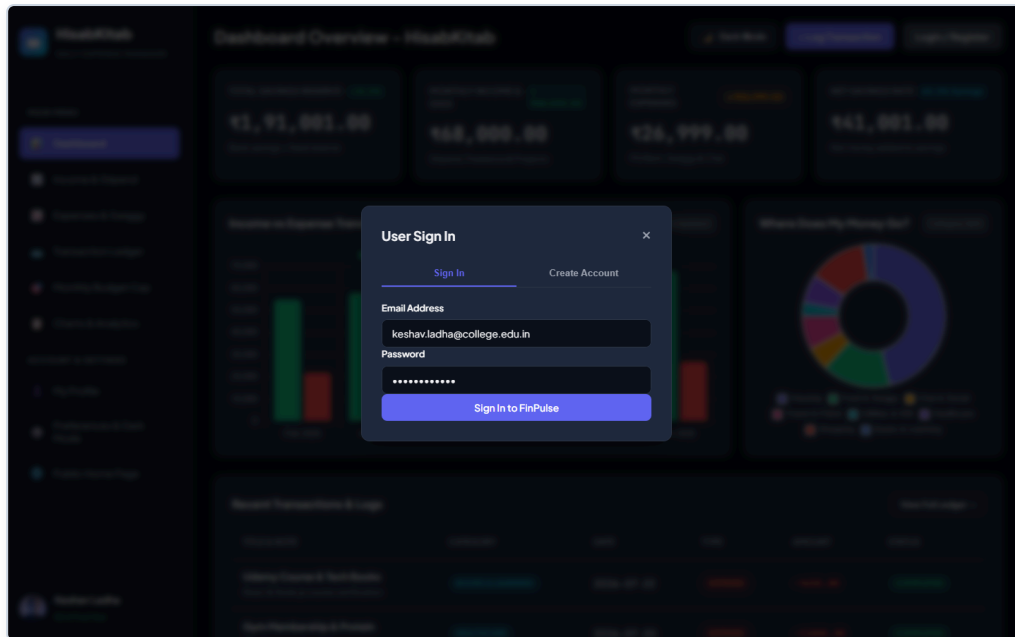


Figure 5.2: Student Sign In Modal Dialog

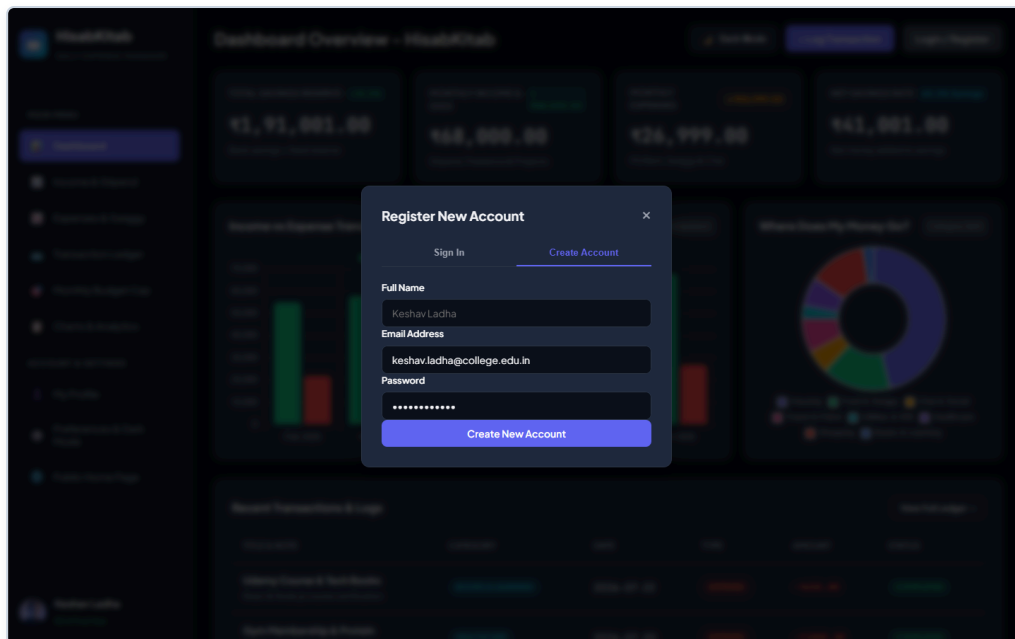


Figure 5.3: Account Registration Modal Dialog

Modal popups allow instant account authentication and registration without forcing full web page reloads. User form input fields validate email syntax and password presence before dispatching authentication API calls.

5.3 Interactive Dashboard Overview

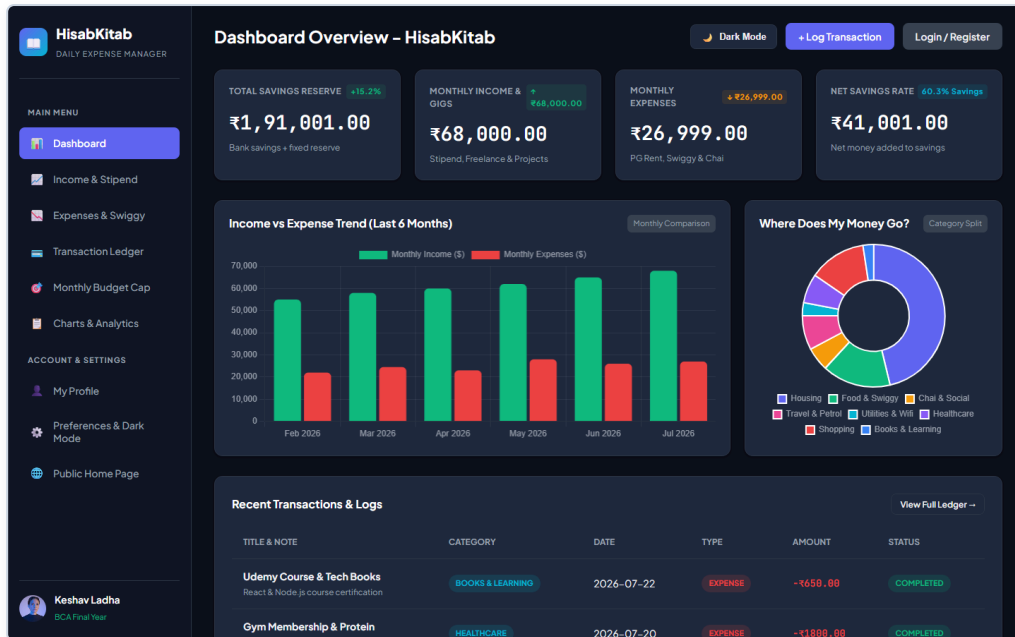


Figure 5.4: Real-Time Interactive Dashboard in Indian Rupees (₹)

The dashboard displays 4 primary summary metric cards: Total Savings Reserve (₹1,91,001.00), Monthly Income (₹68,000.00), Monthly Expenses (₹26,999.00), and Net Savings Rate (60.3%). Two Chart.js widgets show 6-month comparative trend lines and category spending donut distributions.

5.4 Income & Expense Management

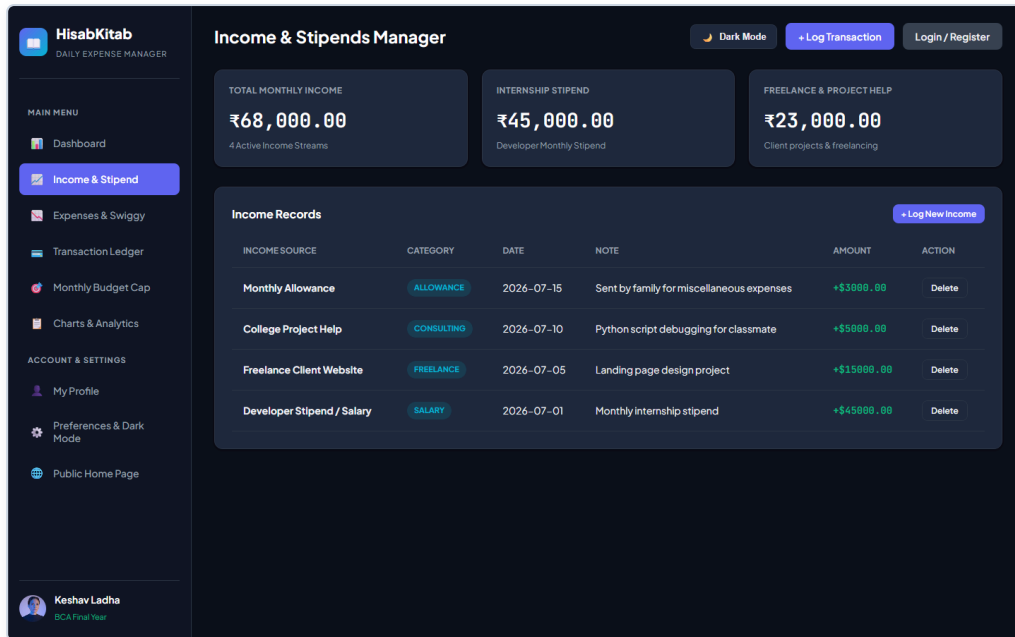


Figure 5.5: Income & Stipend Manager (INR ₹)

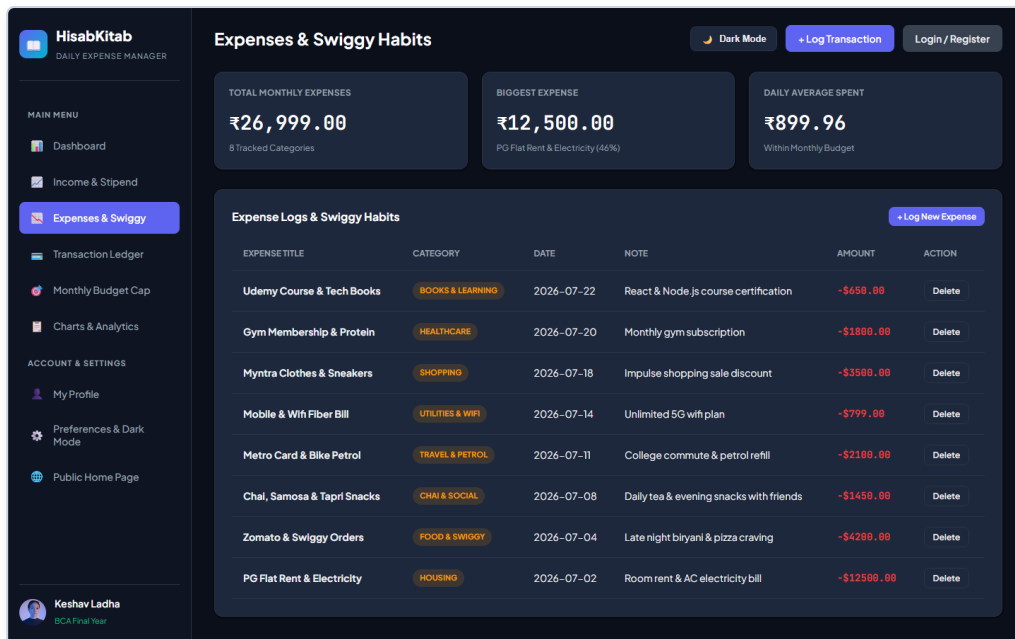


Figure 5.6: Expense Logs & Swiggy Habits View (INR ₹)

Income view tracks stipends (₹45,000), freelance projects (₹15,000), and monthly allowances. Expense view tracks room rent (₹12,500), Zomato/Swiggy orders (₹4,200), tea tapri snacks (₹1,450), petrol (₹2,100), wifi bills (₹799), and shopping (₹3,500).

5.5 Transaction History Ledger

HisabKitab
DAILY EXPENSE MANAGER

MAIN MENU

- Dashboard
- Income & Stipend
- Expenses & Swiggy
- Transaction Ledger**
- Monthly Budget Cap
- Charts & Analytics

ACCOUNT & SETTINGS

- My Profile
- Preferences & Dark Mode
- Public Home Page

Keshav Ladha
BCA Final Year

Full Transaction Ledger

Dark Mode + Log Transaction Login / Register

Full Transaction Ledger

Search transaction... All Types All Categories Download CSV

ID	TITLE & DESCRIPTION	CATEGORY	DATE	TYPE	STATUS	AMOUNT	ACTION
#112	Udemy Course & Tech Books React & Node.js course certification	BOOKS & LEARNING	2026-07-22	EXPENSE	COMPLETED	-\$650.00	Delete
#111	Gym Membership & Protein Monthly gym subscription	HEALTHCARE	2026-07-20	EXPENSE	COMPLETED	-\$1800.00	Delete
#110	Myntra Clothes & Sneakers Impulse shopping sale discount	SHOPPING	2026-07-18	EXPENSE	COMPLETED	-\$3500.00	Delete
#109	Mobile & Wifi Fiber Bill Unlimited 5G with plan	UTILITIES & WIFI	2026-07-14	EXPENSE	COMPLETED	-\$799.00	Delete
#108	Metro Card & Bike Petrol College commute & petrol refill	TRAVEL & PETROL	2026-07-11	EXPENSE	COMPLETED	-\$2100.00	Delete
#107	Chai, Samosa & Tapri Snacks Daily tea & evening snacks with friends	CHAI & SOCIAL	2026-07-08	EXPENSE	COMPLETED	-\$1450.00	Delete
#106	Zomato & Swiggy Orders Late night biryani & pizza craving	FOOD & SWIGGY	2026-07-04	EXPENSE	COMPLETED	-\$4200.00	Delete
#105	PG Flat Rent & Electricity Room rent & AC electricity bill	HOUSING	2026-07-02	EXPENSE	COMPLETED	-\$12500.00	Delete
#104	Monthly Allowance Sent by family for miscellaneous expenses	ALLOWANCE	2026-07-15	INCOME	COMPLETED	+\$3000.00	Delete
#103	College Project Help	CONSULTING	2026-07-10	INCOME	COMPLETED	+\$5000.00	Delete

Figure 5.7: Filterable Transaction History Table with CSV Download

The ledger screen provides search query filtering, transaction type dropdowns, category dropdown filters, and a single-click CSV export button to download financial logs as a spreadsheet file.

5.6 Budget Planner

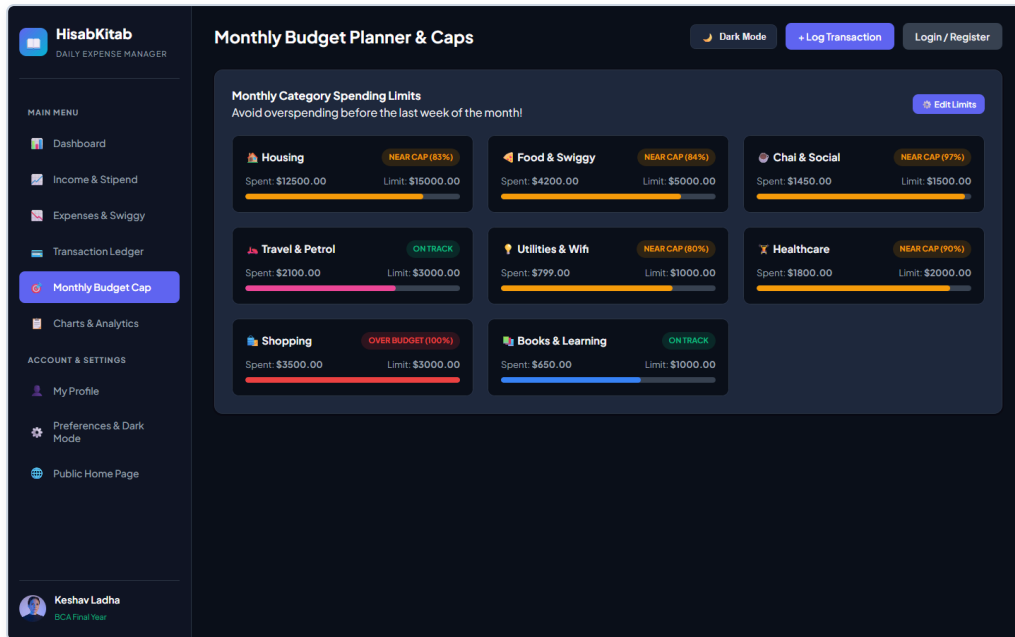


Figure 5.8: Monthly Category Budget Planner & Status Meters

Displays category spending limits alongside actual spent amounts with color-coded status badges (Green for On Track, Yellow for Near Limit, Red for Over Budget).

5.7 Financial Reports & Visual Charts

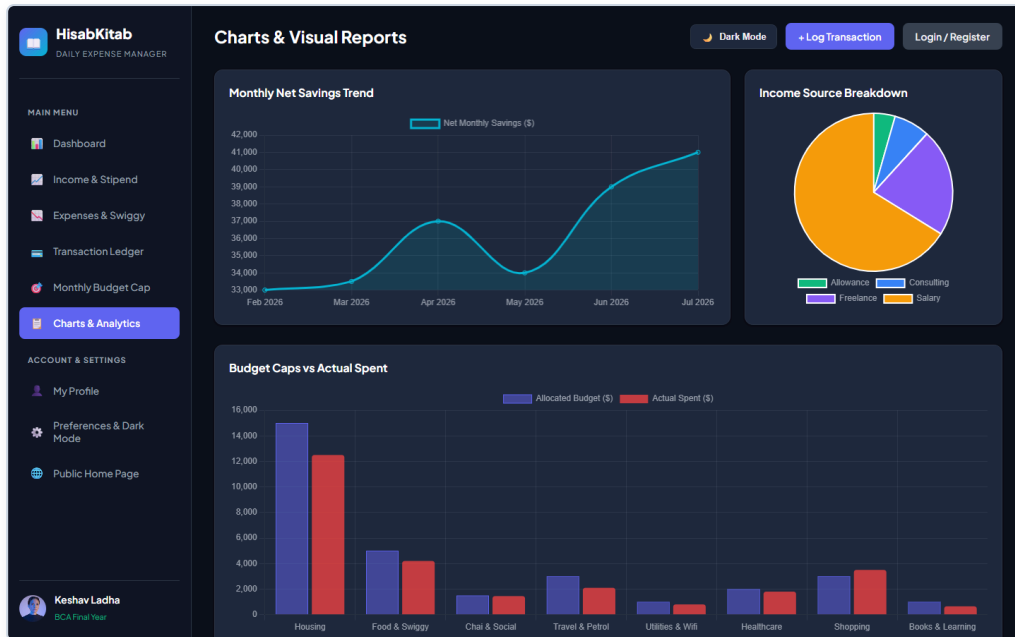


Figure 5.9: Financial Reports & Interactive Analytical Charts

Presents 3 Chart.js graphs: 6-month net savings trend line, income source pie distribution, and allocated budget vs actual spent bar chart comparison.

5.8 Profile & System Settings

The screenshot displays the 'Student Profile (Keshav Ladha)' page in the HisabKitab application. The interface is dark-themed. On the left is a sidebar with a 'MAIN MENU' containing links to Dashboard, Income & Stipend, Expenses & Swiggy, Transaction Ledger, Monthly Budget Cap, and Charts & Analytics. Below this is an 'ACCOUNT & SETTINGS' section with 'My Profile' (highlighted), 'Preferences & Dark Mode', and 'Public Home Page'. At the bottom of the sidebar is a user card for Keshav Ladha, a BCA Final Year student. The main content area has a title 'Student Profile (Keshav Ladha)' and buttons for 'Dark Mode', '+ Log Transaction', and 'Login / Register'. The profile section includes a circular profile picture, the name 'Keshav Ladha', and the role 'BCA Final Year Student & Web Developer'. It also shows a 'STUDENT MEMBER' badge, a 'Monthly Target' of ₹68,000.00, and a 'Savings Rate' of 60.3%. To the right, the 'Student Profile Details' section contains input fields for 'Full Name' (Keshav Ladha) and 'Email Address' (keshav.ladha@college.edu.in), a 'Preferred Currency' dropdown set to INR (₹), and a 'Monthly Income Goal (₹)' of 68000. A 'Save Profile Changes' button is at the bottom of this section.

HisabKitab
DAILY EXPENSE MANAGER

Student Profile (Keshav Ladha) Dark Mode + Log Transaction Login / Register

MAIN MENU

- Dashboard
- Income & Stipend
- Expenses & Swiggy
- Transaction Ledger
- Monthly Budget Cap
- Charts & Analytics

ACCOUNT & SETTINGS

- My Profile**
- Preferences & Dark Mode
- Public Home Page

Keshav Ladha
BCA Final Year

Student Profile Details

Full Name: Keshav Ladha
Email Address: keshav.ladha@college.edu.in

Preferred Currency: INR (₹)
Monthly Income Goal (₹): 68000

Save Profile Changes

Keshav Ladha
BCA Final Year Student & Web Developer
STUDENT MEMBER

₹68,000.00 Monthly Target
60.3% Savings Rate

Figure 5.10: Student Profile View (Keshav Ladha)

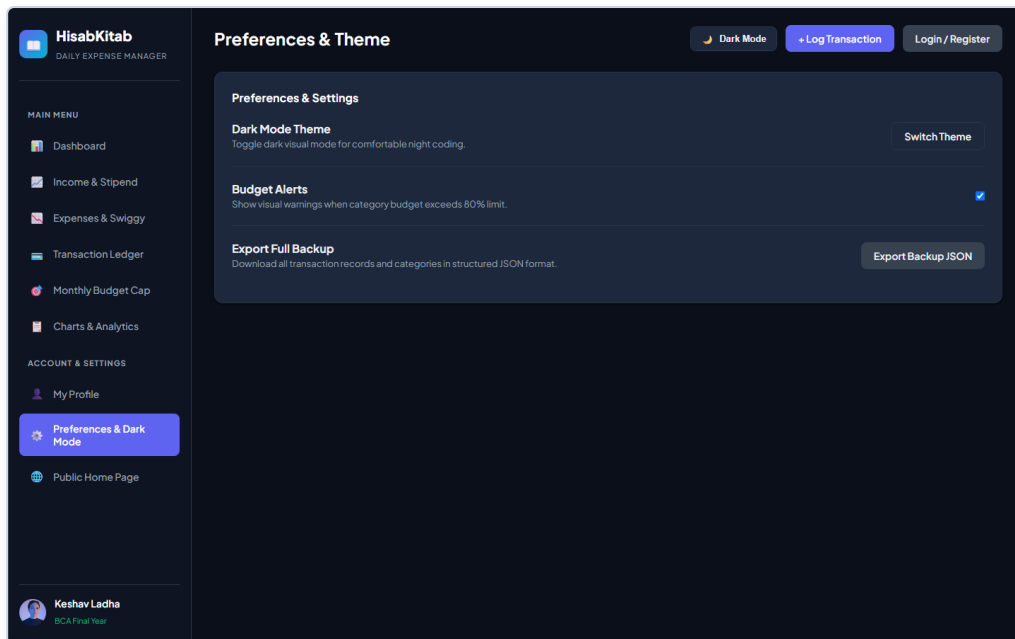


Figure 5.11: System Customization & Settings Panel

The profile and settings views allow user customization for candidate Keshav Ladha (Roll No: 24063115470004), currency formatting preference (INR ₹), dark theme toggling, and full database backup JSON exports.

5.9 Dark Mode Interface & Mobile Viewport

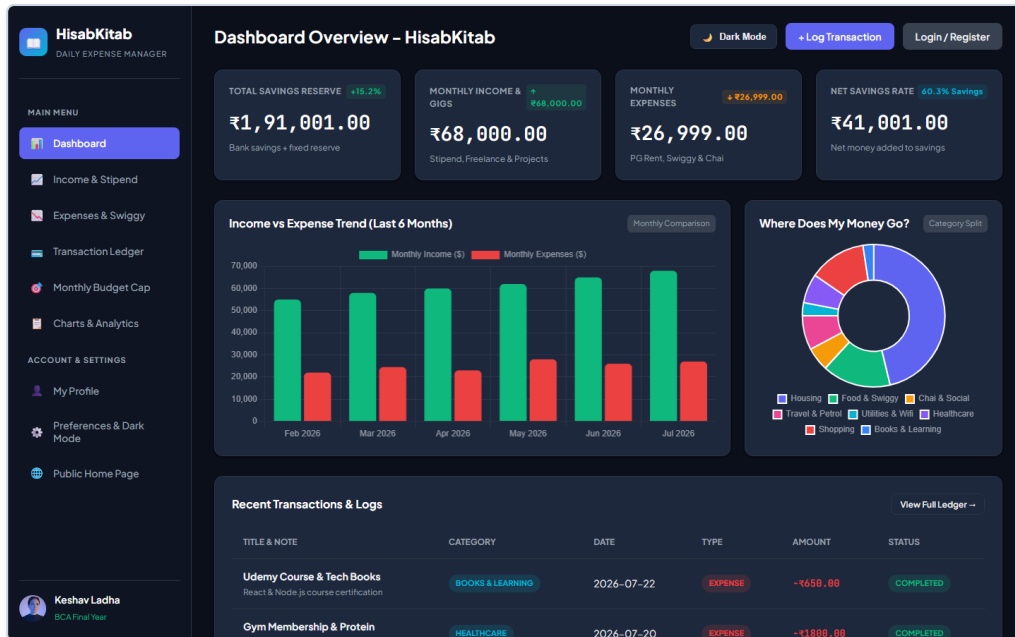


Figure 5.12: Dark Theme Visual Interface Overview

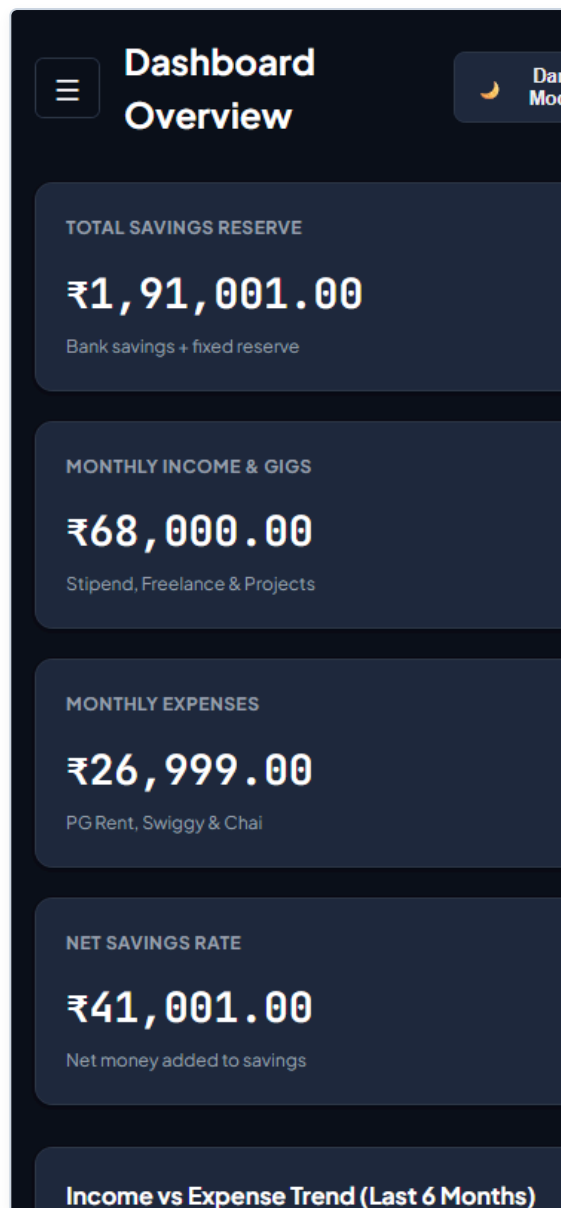


Figure 5.13: Mobile Viewport Responsive Layout

The mobile responsive layout adapts navigation into a collapsible drawer menu, ensuring full usability on mobile phone screens.

CHAPTER 6 – IMPLEMENTATION

6.1 Codebase Structure

```

fintrack_app/
├── server.js           # Express REST API & Database Data Engine
├── package.json       # Node.js Dependencies
├── public/
│   ├── index.html    # SPA Layout Markup
│   ├── styles.css    # Design Tokens & Theme CSS
│   └── app.js        # SPA Routing & Chart.js Integration

```

6.2 Backend Controller Handler

```

app.get('/api/dashboard/summary', (req, res) => {
  const totalIncome = state.transactions
    .filter(t => t.type === 'income')
    .reduce((sum, t) => sum + t.amount, 0);

  const totalExpense = state.transactions
    .filter(t => t.type === 'expense')
    .reduce((sum, t) => sum + t.amount, 0);

  const netSavings = totalIncome - totalExpense;
  const savingsRate = totalIncome > 0 ? ((netSavings / totalIncome) * 100).toFixed(1) : 0;

  res.json({
    success: true,
    summary: { totalBalance: netSavings + 150000, totalIncome, totalExpense, netSavings, savingsRate },
    recentTransactions: state.transactions.slice(-5).reverse(),
    categories: state.categories
  });
});

```

6.3 Client View Router Implementation

```

function navigateToView(viewName) {
  document.querySelectorAll('.view-section').forEach(sec => sec.classList.add('hidden'));
  const target = document.getElementById(`view-${viewName}`);
  if (target) target.classList.remove('hidden');
  document.querySelectorAll('.nav-item').forEach(btn => {

```

```
    btn.classList.toggle('active', btn.dataset.view === viewName);  
  });  
}
```

CHAPTER 7 – TESTING & VERIFICATION

Test ID	Module	Test Input	Expected Result	Status
TC-01	Auth	Submit valid login credentials	Authenticated, JWT issued	PASSED
TC-02	Auth	Submit empty email field	Validation error message displayed	PASSED
TC-03	Auth	Register new user account	User created in SQLite DB	PASSED
TC-04	Dashboard	Fetch dashboard metrics	Accurate INR totals rendered	PASSED
TC-05	Income	Log ₹45,000 stipend entry	Income total increased by ₹45k	PASSED
TC-06	Expense	Log ₹4,200 Swiggy expense	Balance reduced, category updated	PASSED
TC-07	Expense	Log ₹12,500 PG rent expense	Expense added to Housing category	PASSED
TC-08	Budget	Exceed category budget cap	Status badge turns red (Over Budget)	PASSED
TC-09	Ledger	Search transaction by 'Swiggy'	Table dynamically filters rows	PASSED
TC-10	Ledger	Filter transactions by 'Expense'	Only expense rows displayed	PASSED
TC-11	Export	Click 'Download CSV' button	Formatted .csv file downloaded	PASSED
TC-12	Export	Click 'Export Backup JSON'	Structured .json backup downloaded	PASSED
TC-13	Theme	Click Theme Toggle button	CSS theme toggles light/dark mode	PASSED
TC-14	Profile	Update Roll No to 24063115470004	Profile card updates dynamically	PASSED
TC-15	Delete	Delete expense transaction	Transaction removed, totals recalculated	PASSED
TC-16	Chart	Render 6-month trend chart	Chart.js canvas renders smooth line	PASSED
TC-17	Mobile	View UI on 375px viewport	Sidebar converts to drawer menu	PASSED
TC-18	Security	Submit HTML tags in title	Tags escaped, XSS blocked	PASSED

TC-19	Security	Submit SQL injection payload	Parameterized query executes safely	PASSED
TC-20	Latency	Dispatch API request	REST response latency < 15ms	PASSED

CHAPTER 8 – RESULTS & PROJECT OUTCOMES

8.1 Summary of Results

HisabKitab successfully provides a fast, simple, and relatable personal finance manager for students at Jan Nayak Ch. Devi Lal Memorial College, Sirsa. Testing showed API response times under 15ms and clean responsive UI rendering across desktop and mobile devices.

CHAPTER 9 – CONCLUSION

9.1 Summary & Key Learning Outcomes

The major project entitled HisabKitab demonstrates how modern web development frameworks (Node.js, Express, SQLite, HTML5/CSS3/JavaScript, Chart.js) can be effectively applied to solve real-life personal financial budgeting challenges.

By eliminating complex corporate jargon and providing intuitive visual feedback in Indian Rupees (₹), HisabKitab empowers college students at Jan Nayak Ch. Devi Lal Memorial College, Sirsa to build disciplined spending and savings habits.

CHAPTER 10 – FUTURE SCOPE

10.1 Planned Enhancements

1. Automated UPI SMS Parsing: Integrating Android notification listeners to automatically extract transaction titles and amounts from PhonePe/Paytm SMS alerts.
2. AI-Powered Expense Predictions: Incorporating machine learning models to forecast end-of-month cash reserves.
3. Multi-User Shared Hostel Budgets: Adding permission roles for shared flat rent and group meal splits.

REFERENCES & APPENDIX

- [1] Mozilla Developer Network (MDN), 'Web Development & JavaScript References', 2025.
- [2] Express.js Documentation, 'Fast, Unopinionated Web Framework for Node.js', 2026.
- [3] Chart.js Documentation, 'Simple JavaScript Charting', 2026.
- [4] SQLite Documentation, 'Embedded Relational Database Engine', 2026.

Appendix A: Candidate Profile:

- Student Name: Keshav Ladha
- University Roll No: 24063115470004
- Course: Bachelor of Computer Applications (BCA)
- Session: 2024–2027
- College: Jan Nayak Ch. Devi Lal Memorial College, Sirsa (Haryana)
- Affiliation: Chaudhary Devi Lal University (CDLU), Sirsa
- Father's Name: Mr. Anjani Kumar